Suppose you have an N point signal, and need to calculate the *real DFT* by means of the *Complex DFT* (such as by using the FFT algorithm). First, move the N point signal into the real part of the complex DFT's time domain, and then set all of the samples in the imaginary part to *zero*. Calculation of the complex DFT results in a real and an imaginary signal in the frequency domain, each composed of N points. Samples 0 through $N/2$ of these signals correspond to the real DFT's spectrum.

As discussed in Chapter 10, the DFT's frequency domain is periodic when the negative frequencies are included (see Fig. 10-9). The choice of a single period is arbitrary; it can be chosen between -1.0 and 0, -0.5 and 0.5, 0 and 1.0, or any other one unit interval referenced to the sampling rate. The complex DFT's frequency spectrum includes the negative frequencies in the 0 to 1.0 arrangement. In other words, one full period stretches from sample 0 to sample $N-1$, corresponding with 0 to 1.0 times the sampling rate. The positive frequencies sit between sample 0 and $N/2$, corresponding with 0 to 0.5. The other samples, between $N/2+1$ and $N-1$, contain the negative frequency values (which are usually ignored).

Calculating a *real Inverse DFT* using a *complex Inverse DFT* is slightly harder. This is because you need to insure that the negative frequencies are loaded in the proper format. Remember, points 0 through $N/2$ in the complex DFT are the same as in the real DFT, for both the real and the imaginary parts. For the real part, point $N/2+1$ is the same as point $N/2-1$, point $N/2+2$ is the same as point $N/2-2$, etc. This continues to point $N-1$ being the same as point 1. The same basic pattern is used for the imaginary part, except the sign is changed. That is, point $N/2+1$ is the negative of point $N/2-1$, point $N/2+2$ is the negative of point $N/2-2$, etc. Notice that samples 0 and $N/2$ do not have a matching point in this duplication scheme. Use Fig. 10-9 as a guide to understanding this symmetry. In practice, you load the real DFT's frequency spectrum into samples 0 to $N/2$ of the complex DFT's arrays, and then use a subroutine to generate the negative frequencies between samples $N/2+1$ and $N-1$. Table 12-1 shows such a program. To check that the proper symmetry is present, after taking the inverse FFT, look at the imaginary part of the time domain. It will contain all zeros if everything is correct (except for a few parts-per-million of noise, using single precision calculations).

```
6000 'NEGATIVE FREQUENCY GENERATION
6010 'This subroutine creates the complex frequency domain from the real frequency domain.
6020 'Upon entry to this subroutine, N% contains the number of points in the signals, and
6030 'REX[ ] and IMX[ ] contain the real frequency domain in samples 0 to N%/2.
6040 'On return, REX[ ] and IMX[ ] contain the complex frequency domain in samples 0 to N%-1.
6050 '
6060 FOR K% = (N%/2+1) TO (N%-1)
6070   REX[K%] =  REX[N%-K%]
6080   IMX[K%] = -IMX[N%-K%]
6090 NEXT K%
6100 '
6110 RETURN
```

TABLE 12-1